

E – Commerce Sales Analysis

Business Intelligence & Executive Report

Prepared by: SREEISWARYA. S.P

Tools: MySQL | Power BI | DAX

Project Type: E – Commerce Sales Analysis

Reporting Period: 2023 – 2024

Executive Summary:

The E – Commerce business generated approximately 49.38M in sales and 7.38M in profit during the analyzed period, resulting in an overall profit margin of approximately 14.9%. Electronics was the strongest – performing product category in terms of sales and profit. Rajasthan recorded the highest sales among the analyzed states, while Lucknow was the leading city by sales. Customer and payment analysis indicates that revenue is distributed across a broad customer base, while payment methods are relatively evenly distributed. The analysis also highlights the need to monitor discount levels alongside profitability to ensure revenue growth does not come at the expense of margins.

Key Insights:

1. Sales Performance:

Finding: Rajasthan recorded the highest sales among the analyzed states, with approximately 7.2M in revenue.

Business Meaning: This indicates that Rajasthan is an important market for the business and may provide opportunities for further customer acquisition and retention.

2. City Performance:

Finding: Lucknow was the highest – performing city by sales, generating approximately 7.3M

Business Meaning: The strong performance of Lucknow suggests that the company should evaluate whether similar customer characteristics or product preferences can be replicated in other cities.

3. Sales Trend:

Finding: Monthly sales remained relatively strong throughout the period but showed noticeable fluctuations between individual months.

Business Meaning: Management should investigate the reasons for these monthly fluctuations, including promotional campaigns, product availability, customer demand and seasonal effects.

4. Product Category Performance:

Finding: Electronics was the highest – selling product category, generating approximately 10.2M in sales.

Business Recommendations: The company should maintain strong inventory availability for high – performing electronics products while monitoring margins and discount levels.

5. Top Products:

Finding: Tablet was the highest – selling individual product, generating approximately 2.2M in sales.

Recommendation: Maintain sufficient inventory for high – demand products such as tablets and monitor stock availability to avoid lost sales opportunities.

6. Bottom Products:

Finding: Several products recorded comparatively lower sales than the leading products.

Recommendation: Management should evaluate the demand, pricing, inventory levels and promotional performance of low – selling products before deciding whether to continue, reposition or discontinue them.

7. Discount Analysis:

Finding: Categories with different discount levels show differences in profitability, indicating that discount strategies should be evaluated alongside profit performance.

Recommendation: Management should review discount threshold and evaluate whether higher discounts generates sufficient additional sales to compensate for the reduction in margin.

8. Customer Order Frequency:

Finding: The customer order – frequency analysis identifies a group of repeat customers who place multiple orders during the analyzed period.

Recommendation: Develop targeted loyalty and retention campaigns for repeat customers and consider personalized offers based on purchasing behavior.

9. Payment Analysis:

Finding: Customer orders are distributed relatively evenly across the available payment methods, with no single method overwhelmingly dominating the others.

Recommendation: Maintain all major payment options and monitor transaction success rates, customer preferences and payment – related issues before making changes to payment infrastructure.

Recommendations:

Finding	Recommendation	Business Objective
Rajasthan has highest state sales	Strengthen marketing and retention activities	Increase revenue
Electronics leads category sales	Maintain inventory and optimize promotion	Protect sales
Some products have lower sales	Review pricing, demand and inventory	Improve product performance
Discount levels vary by category	Review discount threshold	Protect margins
Repeat customer generate multiple orders	Introduce loyalty strategies	Increase retention
Payment methods are relatively balanced	Maintain multiple payment options	Improve customer convenience

Risks and Limitations:

1. Dataset limitations:

The analysis is based on the available historical transaction dataset and may not represent the complete business environment.

2. No marketing data:

The dataset don't have information about:

- Advertising spend
- Campaign performance
- Customer acquisition cost

Therefore we cannot determine whether marketing campaigns caused sales changes.

3. No inventory information:

We cannot determine whether low sales were caused by:

- Low demand
- Stock – outs
- Poor product visibility

Because inventory data isn't available.

4. Discount analysis limitation:

Discount and profit relationships are observational and should not automatically be interpreted as causal.

5. Customer information:

The dataset contains customer IDs but may not include:

- Age
- Gender
- Income
- Location details beyond transaction geography

Therefore deeper customer segmentation isn't possible.

Future Improvements:

1. Customer Segmentation:

Add:

- Age
- Gender
- Location
- Customer lifetime value
- Purchase frequency

Then create:

New / Active / Loyal / At – risk customers

2. Marketing Analysis:

Add:

- Marketing spend
- Campaign ID
- Campaign channel
- Conversion rate
- Customer acquisition cost

Then calculate:

Marketing ROI

3. Inventory Analysis:

Add:

- Stock quantity
- Stock – out dates
- Inventory cost
- Reorder level

Then identify:

Products with high demand but low inventory.

4. Forecasting:

Use historical sales to forecast:

- Future monthly sales

This could later become a Python / Power BI advanced project.

5. Customer lifetime value:

This would make your customer analysis significantly more advanced.

Conclusion:

The analysis provides an overview of e-commerce sales, profitability, customer behavior, product performance and payment preferences. The business demonstrates strong revenue generation, with Electronics emerging as the leading product category and Rajasthan and Lucknow showing strong geographic performance. However, discount strategies and lower – performance products require further evaluation to ensure revenue growth is accompanied by sustainable profitability. Customer retention, inventory planning and targeted promotional strategies represent potential areas for future improvement.